

F´, Open Source, and the Future

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Agenda

- F´ and Open Source
- Benefits of Open Source
- Lessons Learned
- F´ and the Future

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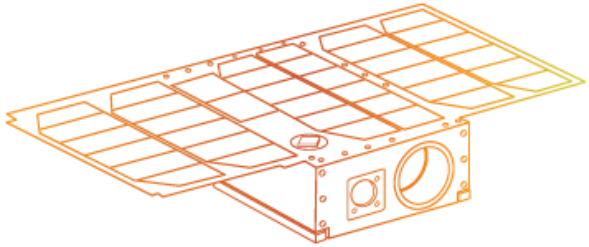
F´ and Open Source

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Framework and Projects

Framework

Standard Components, Development Tools, References



fprime-community

Extras, Add-Ons, Libraries

Project Adaptions

CubeSats, Instruments, Deployables

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Open Sourcing

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https://octodex.github.com/original/

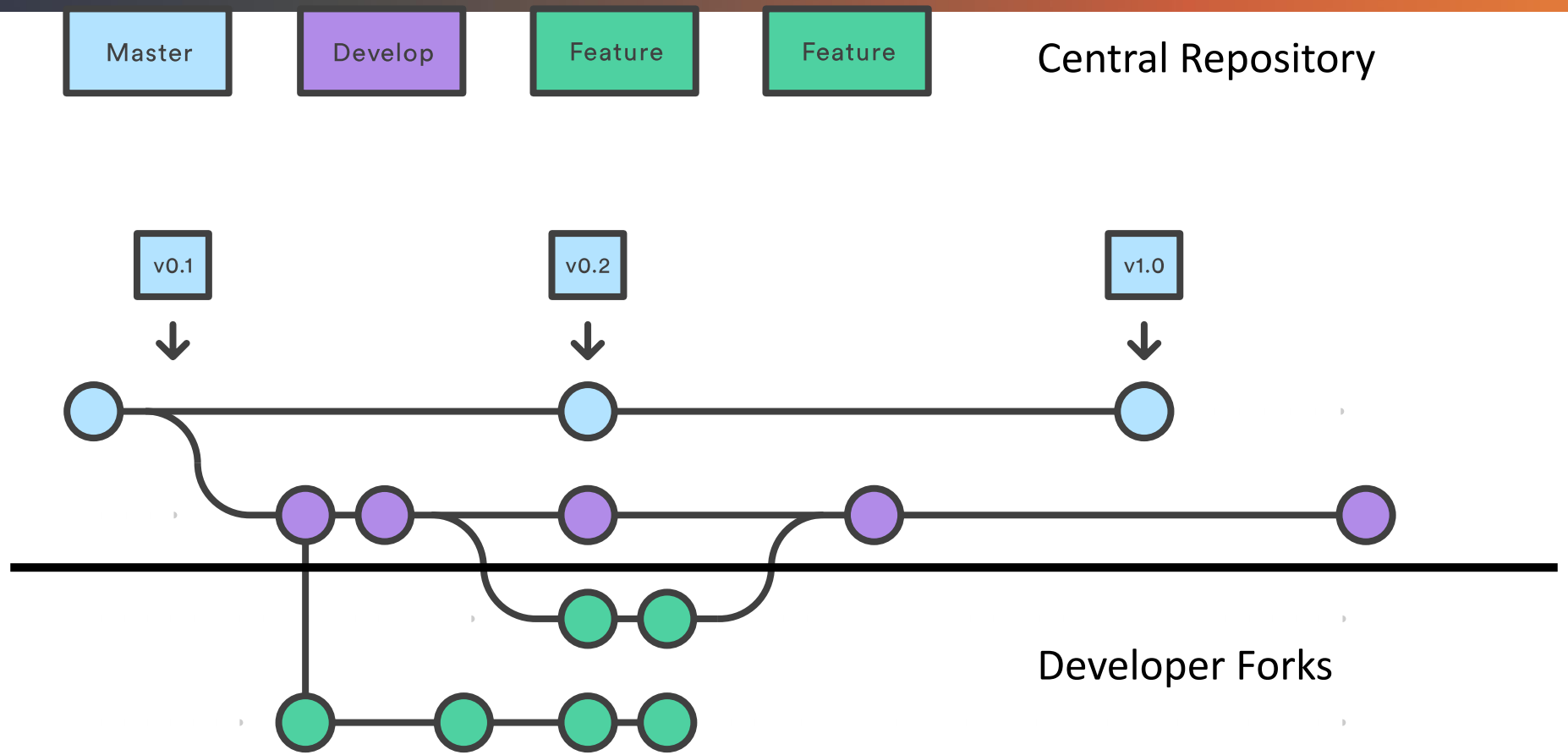
3.5

2013	Internal Development	2017	Open Development	2024

Apache License, Version 2.0

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F´ Open Development Model



https://www.atlassian.com/git/tutorials/comparing-workflows/gitflow-workflow

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Open Tooling and Documentation



Static Analysis



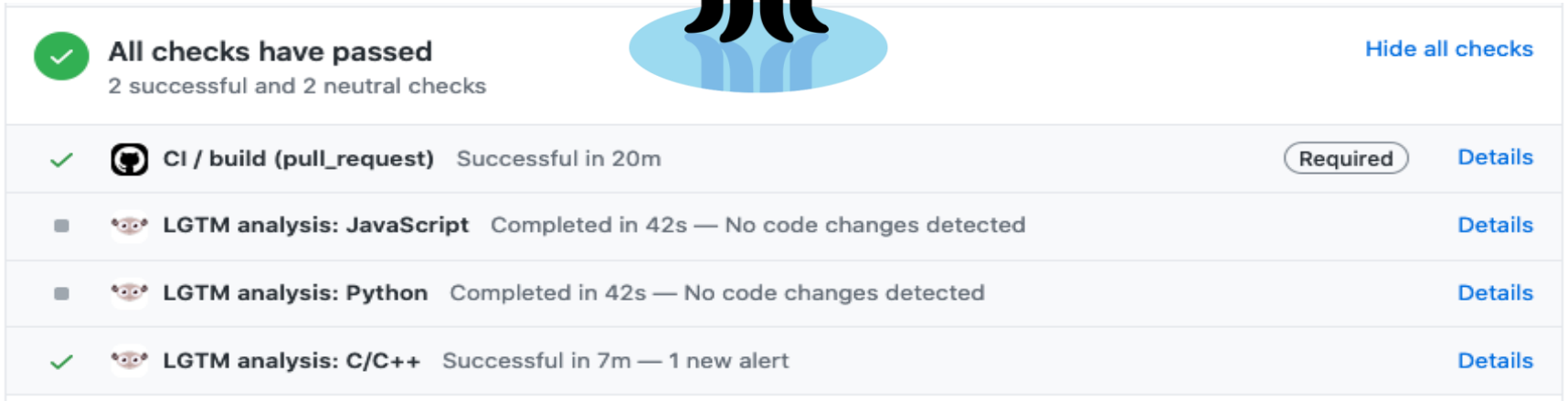
https://www.doxygen.nl/index.html



https://www.sphinx-doc.org/en/master/

Generated Documentation





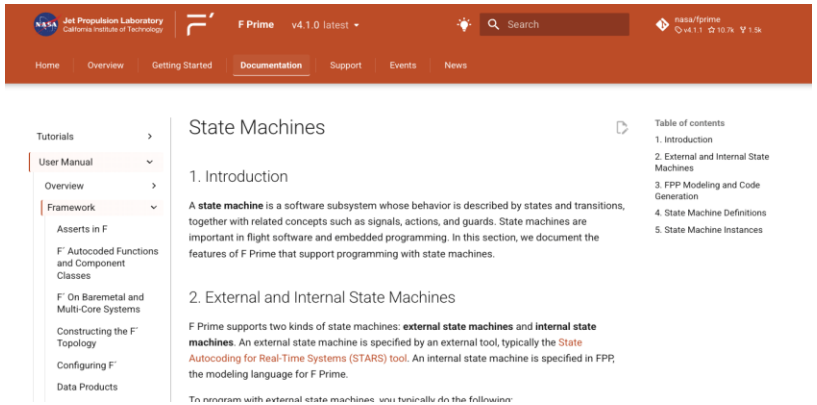
Automatic Pull Request Verification

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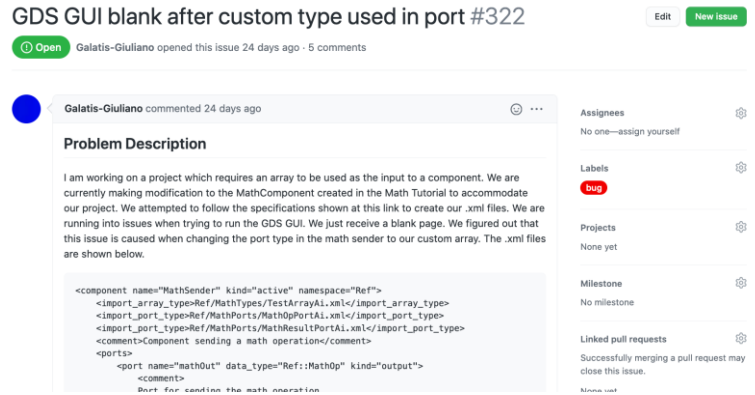
Benefits of Open Source

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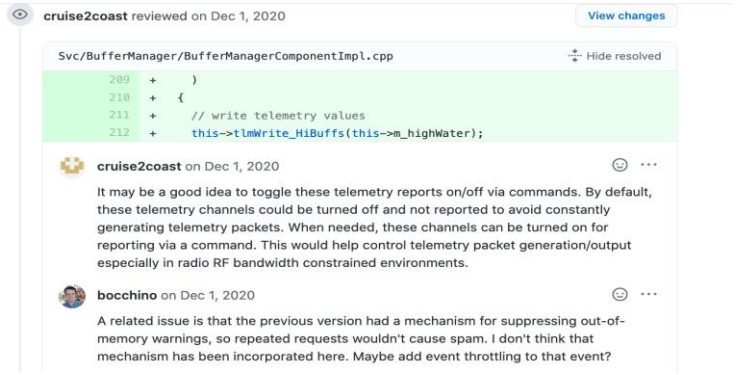
Extended Oversight



Documentation Edits



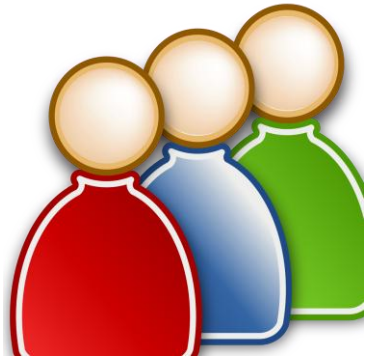
Error Reports



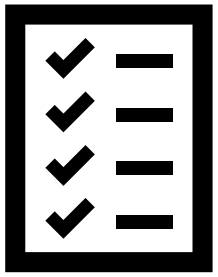
Design and Code Reviews

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Community of Expertise



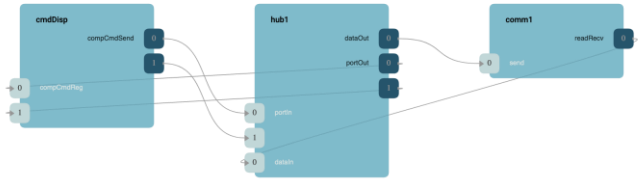
https://commons.wikimedia.org/wiki/File:User-group-square.svg



Best Practices



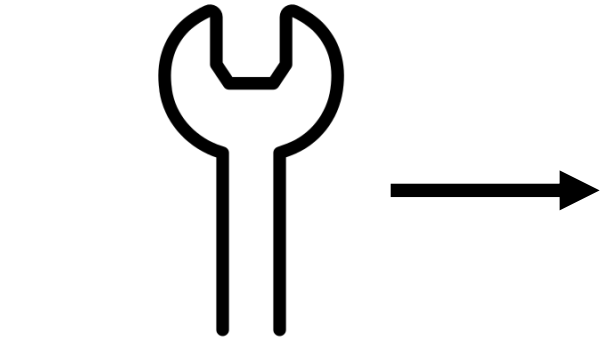
Answers



Recommended Patterns

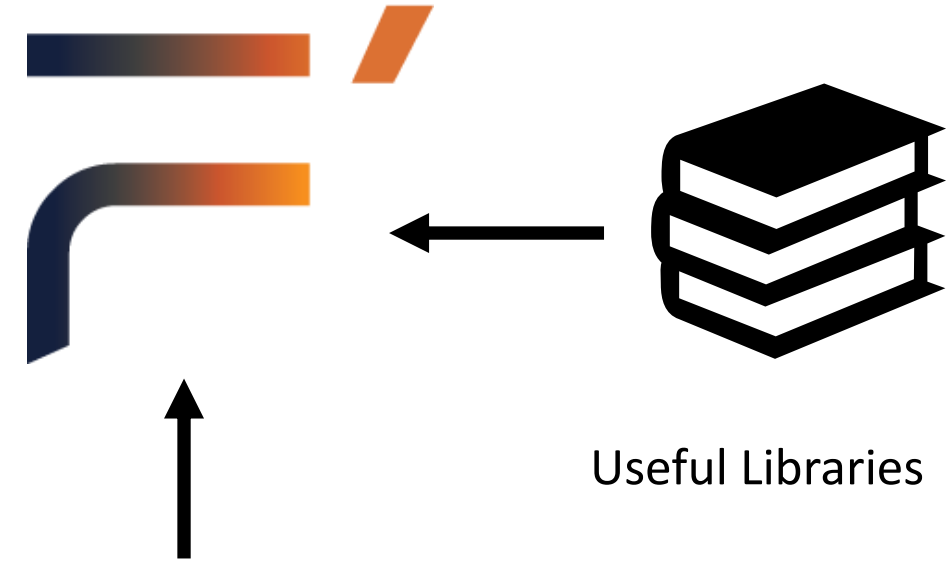
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Ecosystem Surrounding Framework



https://commons.wikimedia.org/wiki/File:OpenMoji-black_1F527.svg

Extended Tooling



Useful Libraries



prjc.com

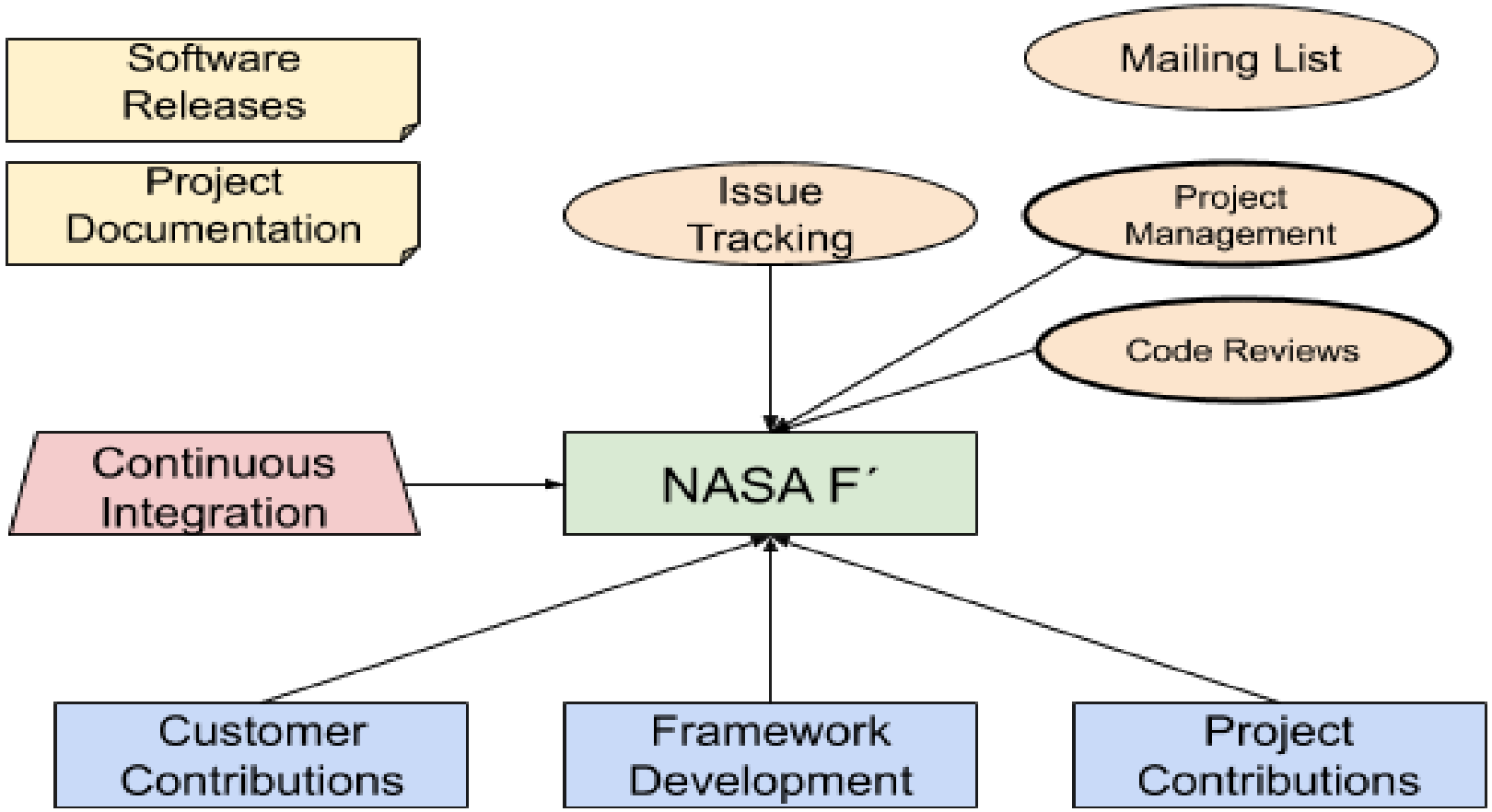
New Use Cases

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Lessons Learned

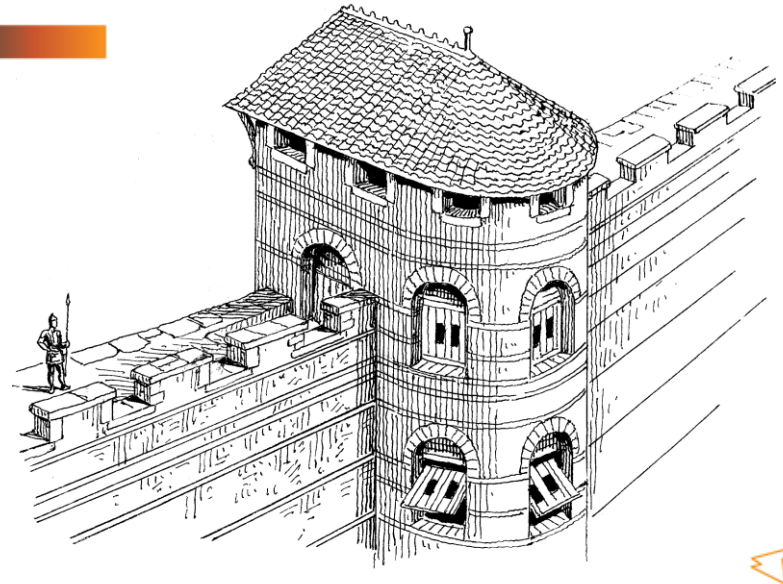
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Embrace Openness

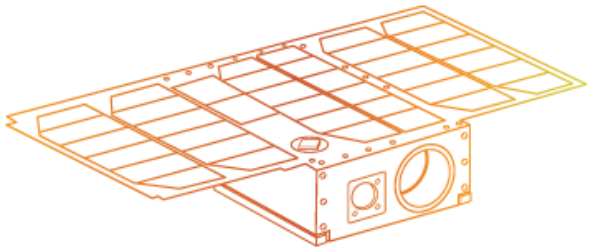


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Separate Product Line from Projects



Pearson Scott Foresman, Public Domain



Budget, Schedules, Short-Cuts

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Leverage Tools and Automation



Free and Open



https://docker.com

Ease Maintenance



Standard Practice



Low Integration Overhead

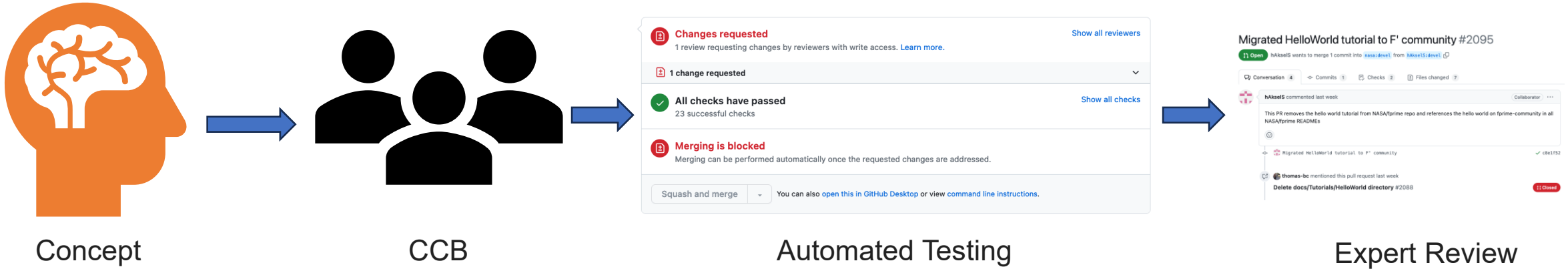


https://github.com/cookiecutter/cookiecutter

Reduce Development

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Accepting Contributions



Concept CCB Automated Testing Expert Review

All contributions to the F´ product line are approved via a change control board (CCB), run through a comprehensive set of automated tests, and finally reviewed by a subject matter expert before merging into the product line.

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Questions?



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Further Information on F Prime

F Prime Website:
https://fprime.jpl.nasa.gov

F Prime Software:
https://github.com/nasa/fprime

F Prime Community:
https://github.com/nasa/fprime/discussions

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